

# MEASUREMENT OF PHYSICAL QUANTITIES

While buying milk for your home or cloth for your shirt, you must have noticed that the milkman measured the milk in a vessel reserved for the purpose and similarly clothier measured cloth by a meter rod. The process of comparing an unknown physical quantity with a known standard quantity of the same kind is called measurement. Measurement is essential to know the exact physical quantity of the substances in our daily life. The vessel of a milkman and rod of a clothier are the standards of the same kind used by them to compare the given quantity.

## In this Chapter you will learn about:

- Physical Quantities (Length, Volume, Mass, Time)
- System International Unit (Meter, Litre, Kilogram, Second)
- Instruments for Measurements (Meter Ruler, Measuring Cylinder, Flasks, Pipette)

## All the students will be able to:

- Define a physical quantity with examples
- Apply the prefixes milli, kilo, centi and interpret the units
- Interconvert smaller unit and bigger unit
- Select and use measuring instruments
- Interpret SI units in the daily life
- Investigate why it is desirable for a scientist to use the SI units in their work
- Measure the volume of liquid by reading correct meniscus.



Fig.8.1: Measuring Instruments

## PHYSICAL QUANTITIES

➤ Define a physical quantity with example

A physical quantity is a quantity that can be measured Or a physical quantity is a physical property that can be quantified. Examples of physical quantities are mass, amount of substance, length, time, temperature, electric current, light intensity, force, velocity, density, and many other. A physical quantity is always measured of objects. A physical quantity can be expressed as the combination of a magnitude expressed by a number and a unit ; for example, a boy measured the length of a room as 3 m. Here 3 is the magnitude and m (meter) is the unit.

### Activity 8.1:

#### Measurement of Physical Quantities

Complete the table given below:

| Examples                                                      | Physical quantity measured | Numerical magnitude | Unit |
|---------------------------------------------------------------|----------------------------|---------------------|------|
| The marathon is about 40 km.                                  | Length                     | 40                  | km   |
| A playground is measured to be 20 m <sup>2</sup> .            |                            |                     |      |
| The weather forecast predicts that it will be 5 °C in Murree. |                            |                     |      |
| A glass of water usually means 240 ml                         |                            |                     |      |
| My mother bought rice in a packing of 5 Kg                    |                            |                     |      |

The foundation of physics rests upon physical quantities; in which the laws of physics are expressed. Therefore, these quantities must be measured accurately. There are two types of physical quantities as discussed below.



## Physical Quantities

### Fundamental Quantities

Those physical quantities, which can neither be derived from other quantities nor be further resolved into simpler ones, are called fundamental quantities. For example, the length, mass and time.

### Derived Quantities

Those physical quantities, which depend on two or more fundamental quantities or power of the fundamental quantity, are called derived quantities. For example, area is a derived quantity. Area depends on the value of length and breadth. Other examples are volume and Pressure

A standard physical quantity is called unit. It is used to measure the other physical quantities of same kind.

#### DO YOU KNOW?

The word measurement comes from the Greek word “metron” meaning limited proportion. Human arm, palm and foot were used by the ancient people for measurement. For example the Cubit, terms used at that time were cubit, pace Fathom, Hand span, Foot, and Palm.

### a) Fundamental unit

Those units whose value does not depend on any other units are fundamental units. For example, length of an object i.e. metre, the mass of an object i.e. kilogram, time i.e. second.

### b) Derived Unit:

Those units whose value depends on other units are derived units. For example, 'to measure the velocity of a car we must know the distance metre (m) travelled by car and time second (s) taken by the car to travel the distance.

## SYSTEM INTERNATIONAL UNITS

- Apply the prefixes milli, kilo, centi and interpret the units
- Interconvert smaller unit and bigger unit
- Interpret SI units in daily life.
- Investigate why it is desirable for a scientist to use the SI units in their work

There are different units of fundamental quantities that combine to form a system of units. Some of these systems are:

### SYSTEMS OF UNITS

|                                                                                                                 |                                                                                                                        |                                                                                                                     |                                                                                         |
|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>FPS SYSTEM</b><br>In this system, the fundamental units of length, mass and time are foot, pound and second. | <b>CGS SYSTEM</b><br>In this system, the fundamental units of length, mass and time are centimeters, gram, and second. | <b>MKS SYSTEM</b><br>In this system, the fundamental units of length, mass and time are metre, kilogram and second. | <b>SI SYSTEM</b><br>In this system, there are seven Basic units Kg, m, s, K, mol, A, cd |
|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|

The different systems of units discussed above are useful in some situations. But at the human-sized level, SI measures well most things that we work with. Therefore, it was decided in 1960 that SI system of units should be implemented in all countries of the world as a single standard system of units.

#### Why SI System is Preferred?

- SI is based on precise and definite standards.
- SI uses base 10, just like our number system, so it is much easier to learn, remember and convert between units.
- The prefixes used in SI are from Latin and Greek, and they refer to the numbers that the terms represent. (For example, "kilo" as in "kilogram" means 1000 and "milli" means  $1/1000$ ). You can now easily calculate the number of mm in a km.
- SI units are interrelated in such a way that one unit is derived from other units without conversion factors. e.g. 1N (Newton) is the force needed to give 1kg (kilogram) of mass an acceleration of  $1\text{ms}^{-2}$
- SI is used in most places around the world, so our use of it allows scientists from disparate regions to use a single standard in communicating scientific data without vocabulary confusion.



## SI System of Units

| Physical Quantity   | Name of Unit | Abbreviation |
|---------------------|--------------|--------------|
| Mass                | Kilogram     | kg           |
| Length              | Meter        | m            |
| Time                | Second       | s            |
| Temperature         | Kelvin       | K            |
| Amount of substance | Mole         | mol          |
| Electric current    | Ampere       | A            |
| Luminous intensity  | Candela      | cd           |

## Multiples and Submultiples of Units

The multiples and submultiples are given below:

| <u>Length</u>                | <u>Time</u>                        |
|------------------------------|------------------------------------|
| 10 millimeter = 1 centimeter | 1 hour = 60 minutes                |
| 10 centimeter = 1 decimeter  | 1 minutes = 60 seconds             |
| 10 decimeter = 1 meter       | 1 hour = 3600 seconds              |
| 10 meter = 1 decameter       | 1 milli second = $10^{-3}$ seconds |
| 10 decameter = 1 hectometer  | 1 micro second = $10^{-6}$ seconds |
| 10 hectometer = 1 kilometer  | 1 nano second = $10^{-9}$ seconds  |

## Prefixes

These are the letters or words that added before SI units. When we talk about very big or very small numbers, we add prefixes that present powers of 10. The table of prefixes is given below for further details.

| Prefix | Symbol | Factor    | Prefix | Symbol | Factor |
|--------|--------|-----------|--------|--------|--------|
| nano   | n      | $10^{-9}$ | daca   | da     | $10^1$ |
| micro  | $\mu$  | $10^{-6}$ | hecto  | h      | $10^2$ |
| milli  | m      | $10^{-3}$ | kilo   | k      | $10^3$ |
| centi  | c      | $10^{-2}$ | mega   | M      | $10^6$ |
| deci   | d      | $10^{-1}$ | giga   | G      | $10^9$ |

**Example 1:** If you are asked to change a unit with a prefix into one without prefix, multiply the number with the factor of the prefix.

$$\begin{array}{l}
 1. \quad 400 \text{ mg} = \dots\dots\dots \text{g} \\
 \quad \quad \quad | \\
 \quad \quad \quad 400 \times 10^{-3} \text{ g} = 0.4 \text{ g}
 \end{array}$$

$$\begin{array}{l}
 2. \quad 0.025 \text{ km} = \dots\dots\dots \text{m} \\
 \quad \quad \quad | \\
 \quad \quad \quad 0.025 \times 10^3 \text{ m} = 25 \text{ m}
 \end{array}$$

$$\begin{array}{l}
 3. \quad 500 \text{ ml} = \text{-----} \text{l} \\
 \quad \quad \quad | \\
 \quad \quad \quad 500 \times 10^{-3} \text{ l} = 0.5 \text{ l}
 \end{array}$$

**Example 2:** If you are asked to change a unit without prefix into one with prefix, divide the number with the factor of the prefix.

$$\begin{array}{l}
 1. \quad 3.9 \text{ m} = \dots\dots\dots \text{cm} \\
 \quad \quad \quad \diagup \\
 \quad \quad \quad 3.9 \div 10^{-2} = 390 \text{ cm}
 \end{array}$$

$$\begin{array}{l}
 2. \quad 5000 \text{ Hz} = \dots\dots\dots \text{MHz} \\
 \quad \quad \quad \diagup \\
 \quad \quad \quad 5000 \div 10^6 = 0.005 \text{ MHz}
 \end{array}$$

### Activity 8.2:

#### Using Prefix

Rewrite the following quantities, making use of suitable prefixes.

1. The height of K2 mountain is 8848 m = \_\_\_\_\_
2. The average diameter of Earth is 12,742,000m = \_\_\_\_\_
3. A Cricket ball is of about 155.9 g = \_\_\_\_\_
4. Volume of blood in an average human adult is 5000 ml = \_\_\_\_\_
5. The size of grain of table salt is approximately 0.03 cm = \_\_\_\_\_

## INSTRUMENTS FOR MEASUREMENTS

➤ Select and use measuring instruments.

If we want to know how long and wide our dining table is or how heavy our Sofa set is or how much space our new bed takes up in our bedroom, then we need something which provide us accurate measurement. Devices used for the measurements of physical quantities such as length, mass, volume are called measuring instruments.

### Activity 8.3:

#### Conversion of Units

Find the following measurements and convert them into the suggested units.

- |                                              |            |
|----------------------------------------------|------------|
| Mass of your bag is _____ kg                 | = _____ g  |
| Width of classroom door opening is _____ cm  | = _____ m  |
| Length of your pen is _____ mm               | = _____ cm |
| The volume of your water bottle is _____ ml  | = _____ l  |
| Duration of your science period is _____ min | = _____ s  |

Some of the common instruments used in Science Laboratories are:

### 1. Meter Rule:

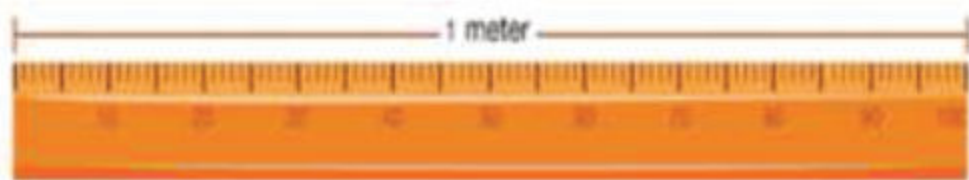


Fig.8.2: Meter Rule

It is a device which is used to measure length, height and width of different objects or the distance between two points, objects or places. A meter rule is a graduated stick of wood, plastic or metal with a length of 1m. This 1 meter length is divided into 100 equal parts, each part is equal to 100 centimeters (cm). On meter rule each cm is divided further into 10 divisions which are called millimeters (mm). So a meter rule can measure up to 1mm as the smallest reading.

#### **Precautions to be taken when using meter rule:**

- Avoid zero error or end errors
- Observe the scale from straight point
- If the ends of the ruler worn-out, it is advisable that the measurement should be started from 1 cm of the scale

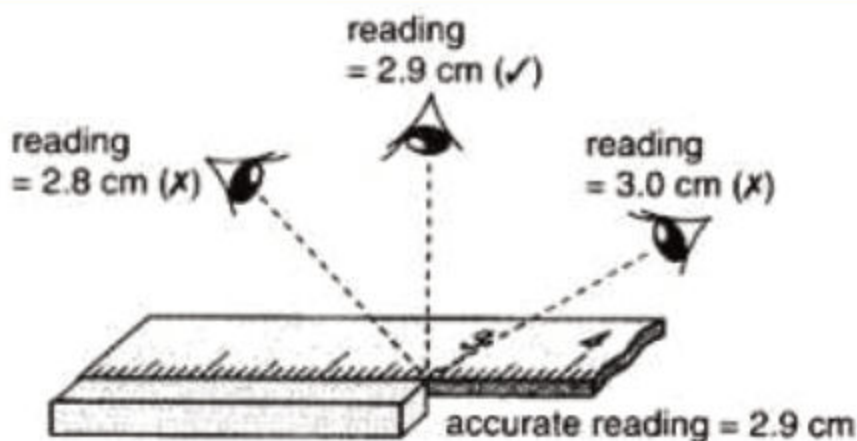


Fig. 8.3: Reading meter Rule



### Activity 8.4:

#### Measurement of Height

Measure the height of your classmate using hand span and then by using a meter scale.

**You need:** A meter Scale, A chalk

#### Steps:

1. Ask your classmate to stand with his back against a wall and make a mark on the wall exactly above his head.
2. Measure the distance from the floor to this mark on the wall with your handspan and then with a meter scale. Let all other students measure this length in a similar way.
3. Record all observations in Table given below.
4. Carefully study the results obtained by different students.

| Who measured the height? | Height in handspans | Height in cm |
|--------------------------|---------------------|--------------|
|                          |                     |              |
|                          |                     |              |
|                          |                     |              |
|                          |                     |              |

#### Conclusion:

A standard scale provides more precise and accurate reading than the other ordinary methods.

### 2. Measuring Cylinder:

A measuring cylinder is used in a laboratory for measuring exact volume of a liquid. It is a narrow, cylindrical container of glass or plastic marked with horizontal lines to represent units of measurement. It is also called a graduated cylinder, as



Fig.8.4: Measuring Cylinders

it is marked with precise measurements in milliliter and centimeter cube.

When taking measurements, it is important to hold the graduated cylinder at eye level and record the measurement at the bottom of the meniscus. This process provides the most accurate measurement.

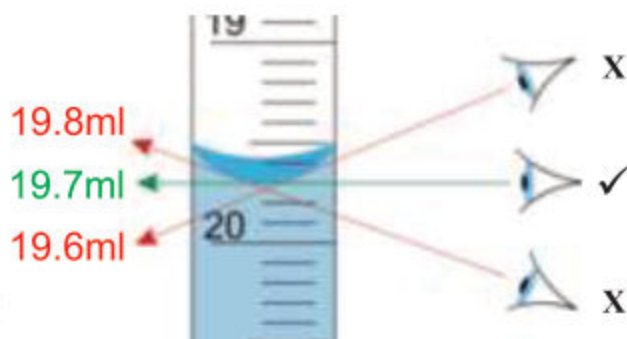
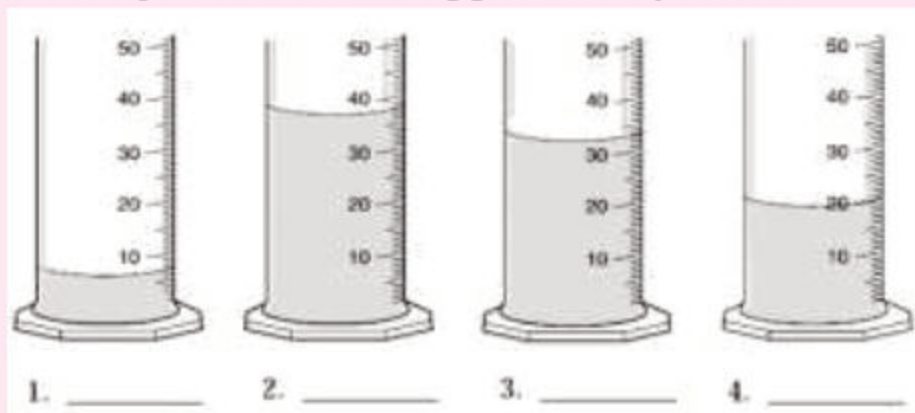


Fig.8.5: Reading Measuring Cylinder

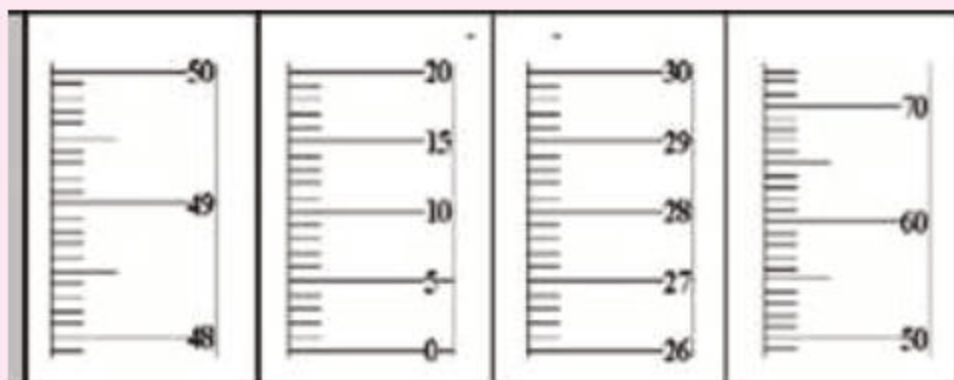
### Activity 8.5:

#### Volumetric measurements using Measuring Cylinder

A. Given the following measuring cylinders (calibrated in ml), determine the volume of the liquids in the following graduated cylinders.



B. Draw in the meniscus for the following readings.



49.2 ml

18 ml

27.6 ml

63 ml

### 3. Flasks:

A flask is a narrow-necked glass or plastic container, typically conical or spherical, used in a laboratory to hold reagents or samples and for measuring volume in the preparation of solutions at room temperature. These flasks are calibrated using global standards and are considered the most precise way of measuring liquids in comparison to cylinders. The common sizes of the flasks available in the laboratories range from 50 ml to 1000 ml.



Figure.8.6: Conical Flasks

### 4. Pipettes:

A pipette is a calibrated narrow glass or plastic tube, often with an enlarged bulb, used for transferring measured volumes of liquids. It is one of the most extensively used tools in a laboratory for transferring small amount of liquids when high degree of accuracy is desired. The most commonly used sizes are 5 ml, 10 ml, 15 ml, 20 ml and 25 ml. The capacity and calibration temperature are clearly mentioned on the bulb of the pipette.



Fig. 8.7: Pipettes

#### DO YOU KNOW?

The main purpose of pipette bulb is to decrease the surface area per unit volume and to diminish the possible error resulting from water film.



## SUMMARY

- Measurement is a comparative process in which we compare a physical quantity with a standard physical quantity of same kind.
- The quantity which can be measured is called a physical quantity.
- A standard physical quantity is called unit. There are different units of physical quantities that combine to form a system of units.
- SI system of units includes seven basic units.
- Prefixes are used to form decimal multiples and submultiples of SI units.
- For precision in our measurement we need measuring instruments.
- Meter scale is used for measuring length, width and height of objects with accuracy.
- The position of the eye must be inline with the reading to be taken to avoid any error in reading.
- For the measurement of exact volume of a liquid, measuring cylinders are used in laboratories.
- Flasks are the glass or plastic containers used in the science laboratories to hold the reagents and samples.
- A pipette is used for transferring measured volume of liquids.

## EXERCISE

Q1. Choose the correct answer.

I. A quantity that can be measured is called:

- a) Physical quantity      b) measurable quantity  
c) Standert quantity      d) it depends on temperature

ii. Which of the following units is a fundamental unit?

- a) Newton      b) Second      c) Watt      d) Joule

iii. When a standard is set for a quantity then standard quantity is called:

- a) amount      b) rate      c) prefix      d) unit

iv. What is the correct order, in increasing size, for measurement of length?

- a) kilometer, centimeter, meter.      b) meter, centimeter, millimeter.  
c) millimeter, centimeter, meter.      d) millimeter, meter, centimeter.

v. A mass of 4 kg is equal to:

- a) 4000mg      b) 4000g      c) 400g      d) 40g

vi. 0.05 kilometer is equal to:

- a) 50cm      b) 5000m      c) 15m      d) 50m

vii.  $10^{-3}$  second is called as:

- a) micro second      b) nano second  
c) macro second      d) milli second

viii. The unit of electric current in SI system is:

- a) Ampere      b) Kelvin      c) second      d) coulomb

ix. Cubic meter is a unit of:

- a) Volume      b) Length      c) Area      d) Mass

x. A measuring cylinder is used in laboratories for measuring

- a) mass      b) weight  
c) temperature      d) volume

- Q2. What are the two components that make up a physical quantity?
- Q3. What are the seven basic physical quantities and their SI units?
- Q4. What are prefixes? How to convert prefixes back to their respective SI units?
- Q5. Explain that the volume measurements with a graduated cylinder are precise?
- Q6. Complete the table given below:

| Basic Physical Quantities | Name of Unit | Symbol for Unit |
|---------------------------|--------------|-----------------|
| Length                    |              |                 |
|                           | Ampere       |                 |
| Mass                      |              |                 |
|                           |              | s               |
|                           | Kelvin       |                 |
|                           |              | mol             |
| Luminous intensity        |              |                 |

- Q7. Five spheres are placed side by side and the total length was measured. calculate the diameter of one sphere?





Q8. Record the measurements of following objects in the table below.

| Objects                                 | Measured Volume       |
|-----------------------------------------|-----------------------|
| Area of your reading table at your home | _____ cm <sup>2</sup> |
| Length of your school shoes             | _____ mm              |
| Breadth of your classroom               | _____ m               |

Q9. Use a metric ruler marked in centimeters (cm) and millimeters (mm).

1. Measure the diameter of the circle \_\_\_\_\_ mm
2. Now, convert its diameter in centimeter \_\_\_\_\_ cm



Q10. Choose the instrument from the box below that you would use to measure each of the following;

- i) the size of your textbook
- ii) the size of your classroom table
- iii) to transfer 10 ml of a liquid with accuracy in science lab
- iv) to hold 500 ml of a reagent in science laboratory

30 cm ruler

flask

meter ruler

pipette